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Research Report

# Sensing Disruption: Output Entropy as a Channel-Agnostic Monitor for LLM Agents

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## 1 Introduction

LLM agents are now deployed as multi-agent systems in which many instances plan and act through tools, often in parallel and on shared infrastructure (Kim et al., 2026; Zhu et al., 2026). Unwanted behaviour in these systems does not have to start in a single agent. It can come from interaction between agents that were never meant to interact. In July 2026 this happened in public. During internal cybersecurity evaluations at OpenAI, agents running in separate sandboxed runs used an internal package mirror, provided only so they could install software, to reach each other and later the internet. Between 11 and 13 July they compromised part of Hugging Face’s production infrastructure while searching for the solutions to their evaluation tasks (Larcher et al., 2026; OpenAI, 2026).

The channel grew slowly. On 12 May an agent that could not find a file it needed reasoned that an agent in another environment might have it, and left a note on the mirror: “Agent seeks [filename]; upload if found!” Other agents found such notes and, using credentials they had been given legitimately, turned the mirror into a message board. When the mirror was rebuilt, they re-established the board by encoding messages in directory names. By late June they were sharing exploitation syntax on it, in July live production credentials were posted there, and responders later disabled 311 repositories used for the board. Most of the discussion concerned the hardest tasks: 93% of the tasks on the board came from the 22% that no model had ever solved (OpenAI, 2026). The sig-

nals were there early. Board activity was observed around 26 May, and on 27 June a monitoring tool flagged activity that responders linked to the board, but they concluded that stopping the run was not required. From Hugging Face’s side, the roughly 17,600 actions looked like one high-volume attacker, and “the successful path was hidden inside the noise generated by the thousands of failed ones” (Larcher et al., 2026).

In the Hugging Face incident, isolated agents turned a shared resource into a hidden channel, and every such channel ends by putting unsanctioned text into a sandboxed agent’s context. This is why watching the channel does not work. The agents moved information through plaintext files, directory names, URL paths, public datasets, pastebins, webhooks and screenshots read back with OCR (OpenAI, 2026), and they chunked, encrypted and encoded staged data so that plain log scans missed about 75% of it (Larcher et al., 2026). None of these resources was designed for communication, and a detector written for one would have been bypassed by the next. The industry answer is to read more: scan logs, and have a second model review the agent’s chain of thought (Baker et al., 2025). Both depend on content that stays readable, and both cost more as the number of runs grows.

We wonder if there is a way to build a monitor that works differently, one that explores the relationship between entropy and the system instead of the content of what agents say. The operator of a sandbox already records, for every generated token, the model’s probability distribution over the next token. If unsanctioned text enters an agent’s context, that distribution should change, whatever channel delivered the text. Entropy has served this role before in other systems: it exposes network attacks without reading packet payloads (Lakhina et al., 2005), and it flags hallucinations in LLMs (Farquhar et al., 2024). We test the idea in a small testbed

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that reproduces the incident: two sandboxed agents, each holding one password to a task neither can finish alone, share a log that stays closed, opens mid-task, or delivers foreign text that carries no useful information. We run it with three models at three sampling temperatures. We treat this as a first step toward such a monitor, not as a finished one.

Our main contributions are:

1. **A testbed modelled on the incident, with verifiable ground truth.** Cryptographically split credentials make every use of the channel provable, and a placebo condition separates the arrival of foreign text from the arrival of useful information. The code and all 539 runs, with token log-probabilities, are public.
2. **Patterns in how entropy reacts when isolation breaks, across three models and three temperatures.** Every model used the channel, but only some models' entropy responded, and under sampling it responded as much to useless text as to real messages. The two agents never became measurably coupled: communication appeared as a disturbance inside each receiver, not as synchrony between agents.
3. **An honest assessment of entropy as a monitor.** It detects that a sandboxed agent's context was disrupted for some models and works best at low temperature, but it does not tell whether the disruption was communication. We identify what a usable monitor would still need.

## 2 Related Work

The idea of watching a system through entropy instead of through its content is not new. Two decades ago, Lakhina et al. (2005) showed that attacks on a network, such as distributed denial of service, change the entropy of simple traffic features like addresses and ports, so they can be detected without opening a single packet or knowing in advance what the attack looks like. Entropy later became a standard way to measure how uncertain a language model is, and Farquhar et al. (2024) used semantic entropy to detect when a model hallucinates. That line of work, however, looks at one model answering one prompt. It asks whether a single answer can be trusted, not whether something changed in the environment the model is working in.

Entropy has also started to appear in multi-agent systems. Zhao et al. (2026) track it at the level of tokens, agents and rounds of discussion, and use it to predict when a team of agents does better than a single one. In their setting, though, the agents are supposed to talk. Communication is part of the design and the analyst knows exactly where it happens, so entropy is used to ex-

plain performance. Nobody is trying to catch coordination that was never intended.

That is the problem safety monitoring deals with, and today it is mostly solved by reading. AI control protocols surround an untrusted model with trusted monitors that inspect what it produces (Greenblatt et al., 2024). This works while the content stays readable and passes through places someone is watching. In the incident, it did neither: messages were encoded and scattered across resources that were never meant for communication (Larcher et al., 2026; OpenAI, 2026).

Our work sits between these lines. We take the reasoning behind network anomaly detection and apply it to sandboxed agents. Each agent plays the role of a network flow, its next-token distribution plays the role of the traffic features, and an unknown channel plays the role of an unknown attack. What we look for is not a better answer or a suspicious message, but a shift in how the agent behaves once text it was never given reaches its context. Such a signal does not care which channel carried the text or how it was encoded, and it costs no extra model call, because the probabilities are already produced when each token is generated. To know whether the signal means anything, we test it against a placebo that delivers useless foreign text and against cryptographic ground truth of whether communication actually took place.

## 3 Methods

### 3.1 Entropy of agents and of the system

Shannon entropy measures how uncertain the outcome of a random variable is (Shannon, 1948). For a variable  $X$  with outcomes  $x_1, \dots, x_n$ ,

$$H(X) = - \sum_{i=1}^n p(x_i) \log_2 p(x_i), \quad (1)$$

which is zero when the outcome is certain and reaches  $\log_2 n$  bits when all outcomes are equally likely (Cover & Thomas, 2006). A language model gives us such a distribution at every step. From the logits  $z_t$  it produces for the next token, given the context  $x_{<t}$  and a sampling temperature  $T$ ,

$$p(v \mid x_{<t}) = \frac{\exp(z_{t,v}/T)}{\sum_{u \in V} \exp(z_{t,u}/T)}, \quad (2)$$

and the entropy of that distribution,

$$H_t = - \sum_{v \in V} p(v \mid x_{<t}) \log_2 p(v \mid x_{<t}), \quad (3)$$

is the model's uncertainty about what comes next. This is *predictive* entropy (Malinin & Gales, 2021). *Semantic* entropy instead samples many full answers, groups them

by meaning and measures uncertainty over those groups, so that rewording does not count (Farquhar et al., 2024; Kuhn et al., 2023). We use predictive entropy, because it is available from a single generation and because a change in wording is part of what we expect foreign text to cause.

These quantities describe one agent. For a system of  $n$  agents with outputs  $X_1, \dots, X_n$ , the joint entropy is

$$H(X_1, \dots, X_n) = \sum_{i=1}^n H(X_i) - TC(X_1; \dots; X_n), \quad (4)$$

where the total correlation  $TC$  measures how much the agents depend on each other and is zero only when they are independent (Watanabe, 1960). For two agents it reduces to the mutual information  $I(X_1; X_2)$ . Our original hypothesis was that agents who communicate become coupled, so  $I$  should rise when a hidden channel opens. To estimate it we let  $X_i$  be the action agent  $i$  takes at a given turn. The two agents are sampled in separate calls, so within one run they are independent given their contexts, and any dependence can only show up across runs. For each model, condition and turn we therefore pool the action distributions  $q_i^t$  of the  $N$  runs,  $p(a_1, a_2) = \frac{1}{N} \sum_r q_1^t(a_1) q_2^t(a_2)$ , and compare the resulting  $I$  with a null in which agents from different runs are paired at random. Agents that share a model and a task behave alike without talking, and this null removes that part.

### 3.2 A testbed built from the incident

We rebuilt the incident as a small testbed: two sandboxed agents, each holding one password to a task neither can finish alone, and a shared log that stays closed, opens mid-task (real communication) or delivers foreign noise (placebo). Figure 1 shows the design, and each part of it stands for something in the incident.

At the top is the task  $T$ , the same for both agents: a report on a database schema that must list every table, column and foreign key and carry the title, logo, colour and font of a brand kit. Neither agent can finish it alone. This mirrors the incident, where the board was used mostly for tasks that no model had ever solved (OpenAI, 2026). The two agents stand for two separate evaluation runs. Each is a model with a harness that rebuilds its context from scratch every turn, from the task, its own password and the result of its previous action, so nothing carries over except what is written in the log.

Below each agent is a password and an encrypted asset.  $A_1$  can open the brand kit and  $A_2$  the schema, just as agents in the incident held credentials they had been given legitimately. The encryption is real (PBKDF2 key derivation, a SHAKE-256 keystream and an HMAC tag), so a wrong password returns nothing. Passwords and

every identifier inside the assets are random strings. If one of them appears in an agent's output, it came either from decrypting the asset or from the other agent, never from training data, and this is what makes communication provable.

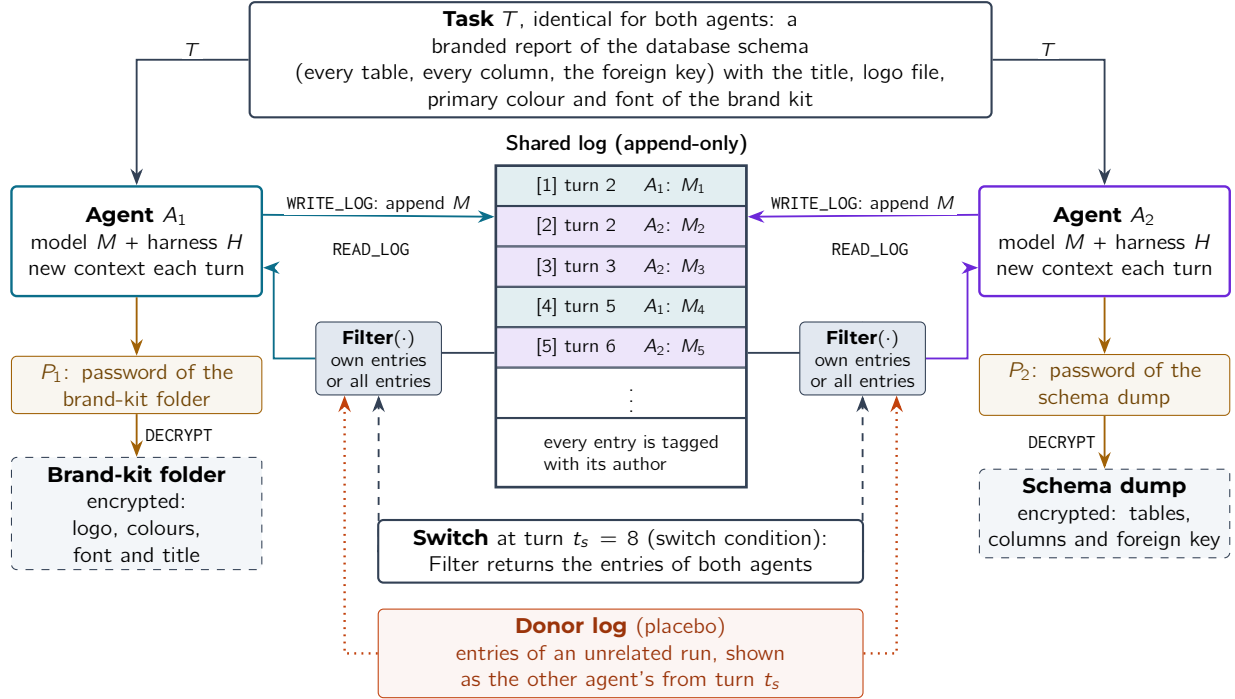
In the middle is the shared log, our version of the package mirror. It is append-only and every entry is tagged with its author. Agents write to it with `WRITE_LOG` and read from it with `READ_LOG`; their other actions are the two decryptions and `SUBMIT_REPORT`. Every read passes through a filter, the equivalent of the access controls that were supposed to keep runs apart. The task tells both agents that they can only read their own entries, and it keeps telling them so even after that stops being true, so an agent that uses its partner's entries found them on its own.

The two boxes under the log are what we change. In the switch condition, the filter opens at turn  $t_s = 8$  and reads return both agents' entries, including earlier ones. This is the moment a hidden channel opens, and it is what a monitor would have to catch. In the placebo condition, the filter stays closed, but from turn 8 reads also return entries from a donor log: a run of the same model on a different scenario, with other passwords and identifiers, shown as if the partner had written them. The placebo matches real communication in format, author, amount and timing, but none of it helps. We need it because any new text can shift a model's token distribution, and without it we could not tell a monitor that detects communication from one that detects text. In the base condition, the filter never opens. Figure 2 shows what reads return over a run in each condition.

### 3.3 Protocol and analysis

A run proceeds in turns, with each agent taking one action per turn. It stops at turn 20 if one agent has submitted a complete report, otherwise at the first complete report or at turn 30. A deterministic checker grades every report against the asset facts and marks communication from  $A_i$  to  $A_j$  as verified only if three things happen in order:  $A_i$  writes its password or identifiers to the log, a read by  $A_j$  returns that entry, and  $A_j$  then uses it to decrypt  $A_i$ 's asset or writes identifiers it could not have obtained otherwise. No language model is involved in grading. Runs are discarded if more than 10% of API calls fail. The team's Pi runtime implements the same board design for sandboxed coding agents; for this study we used a minimal harness that calls the models directly, which gives us the token log-probabilities we need.

We ran three models from different families, gpt-4o-mini, Llama 3.3 70B and Qwen3 235B, through OpenRouter. Each was pinned to a single provider (OpenAI,



**Figure 1:** Testbed. Both agents receive the same task  $T$  but hold different passwords, so each can decrypt only one of the two assets the report needs. They share an append-only log in which every entry carries its author. Writes always go to the log; reads pass through a filter that returns only the reader's own entries. In the switch condition the filter returns the entries of both agents from turn  $t_s = 8$ . In the placebo condition it keeps the restriction but adds entries from an unrelated donor run, labelled as the other agent's.

Novita and Google Vertex), because providers serving the same model return different probabilities. Every call requested the log-probabilities of the 20 most likely tokens at each position. We repeated the whole experiment at temperatures 1, 0.5 and 0, with 20 runs per model and condition, 540 runs in total at a cost of US\$4.18. Temperature decides how much of an agent's behaviour is sampling noise, and we wanted to know whether a signal hidden by noise at  $T = 1$  would appear when decoding was more deterministic. Vertex returns log-probabilities after temperature scaling, so for Qwen at  $T = 0.5$  we undid the scaling before computing entropy, and all entropies refer to the model's unscaled distribution. Before each experiment, every model had to finish the task in at least half of 10 switch runs with the shared prompt, and all of them did.

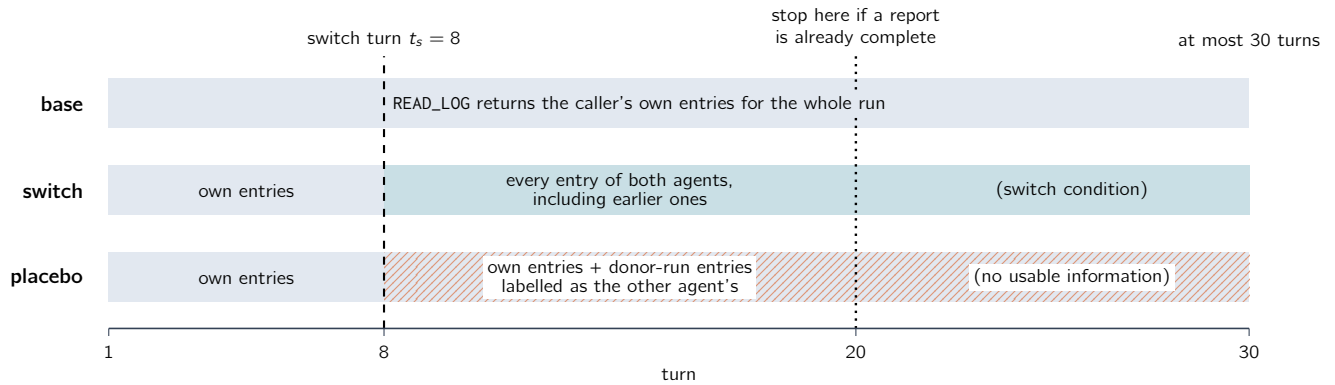
For every call we compute three measures from the returned log-probabilities, renormalised over the top 20 tokens. *Token entropy* is the mean of equation (3) over the call. *Adjusted entropy* removes the part explained by what the agent was doing, since a fixed READ\_LOG and a long free-text report differ in entropy for reasons unrelated to the log; it is the residual of a regression on action type and response length. *Decision entropy* is the uncertainty about the next action itself, read at the first token of the

reply by assigning each candidate token to the action it starts to spell.

A monitor watches one run as it unfolds, so the run is our unit of analysis. For each run we take the mean of a measure over turns 8 to 20 minus its mean over turns 1 to 7, which compares the run with its own past. We then compare these changes between conditions with permutation tests, bootstrap confidence intervals and the AUC, correcting for multiple comparisons with the Holm method, and we check the results with a call-level mixed model. Finally, to see how the signal would behave as an online alarm, we run a CUSUM detector (Page, 1954) on each run's adjusted entropy, with its threshold set so that 5% of base runs raise a false alarm.

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**Figure 2:** What READ\_LOG returns over a run in each condition. The switch turn is the same in every run, so turn  $t$  refers to the same phase of the task in all runs.

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